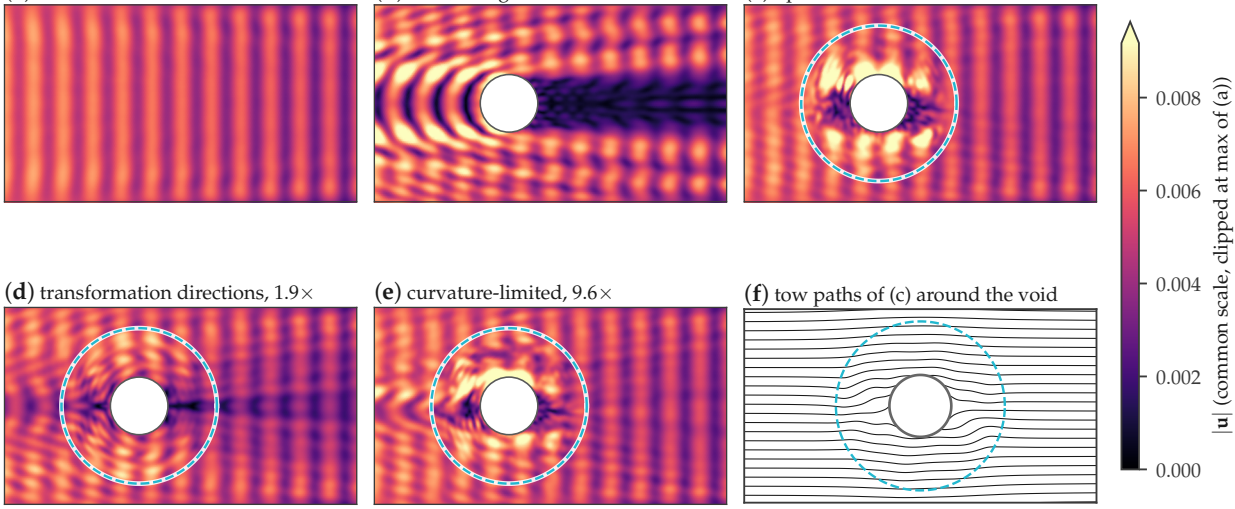

(a) void-free reference
(b) void, straight fibers
(c) optimized shell, 36×
(d) transformation directions, 1.9×
(e) curvature-limited, 9.6×
(f) tow paths of (c) around the void
0.008
0.006
0.004
0.002
0.000
|u| (common scale, clipped at max of (a))